

Neuroscience

This section presents the requirements for programs in:

- **Neuroscience and Mental Health B.Sc. Honours**
- **Neuroscience and Mental Health B.Sc. Major**
- **Neuroscience and Mental Health B.Sc.**
- **Neuroscience and Biology B.Sc. Combined Honours**
- **Minor in Neuroscience and Mental Health**

Program Requirements

Course Categories for B.Sc. Programs

The program descriptions for B.Sc. Combined Honours Neuroscience make use of the course categories defined for all B.Sc. programs (see Academic Regulations for the Bachelor of Science Degree):

- Science Faculty Electives
- Science Continuation Courses
- Free Electives

Neuroscience and Mental Health B.Sc. Honours (20.0 credits)

A. Credits Included in the Major (11.0 credits)

1. 6.0 credits in:	6.0
NEUR 1202 [0.5]	Neuroscience of Mental Health and Psychiatric Conditions
NEUR 1203 [0.5]	Neuroscience of Mental Health and Neurological Conditions
NEUR 2001 [0.5]	Introduction to Research Methods in Neuroscience
NEUR 2002 [0.5]	Introduction to Statistics in Neuroscience
NEUR 2004 [0.5]	Fundamentals of Scientific Writing in Neuroscience
NEUR 2201 [0.5]	Cellular and Molecular Neuroscience
NEUR 2202 [0.5]	Neurodevelopment and Plasticity
NEUR 3001 [0.5]	Data Analysis in Neuroscience I
NEUR 3002 [0.5]	Data Analysis in Neuroscience II
NEUR 3204 [0.5]	Neuropharmacology
NEUR 3206 [0.5]	Sensory and Motor Neuroscience
NEUR 3207 [0.5]	Systems Neuroscience
2. 1.0 credit in:	1.0
BIOL 1103 [0.5]	Foundations of Biology I
BIOL 1104 [0.5]	Foundations of Biology II
3. 1.5 credits from:	1.5
NEUR 3003 [0.5]	Epidemiology in Neuroscience
NEUR 3301 [0.5]	Genetics of Mental Health
NEUR 3303 [0.5]	The Neuroscience of Consciousness
NEUR 3304 [0.5]	Hormones and Behaviour
NEUR 3401 [0.5]	Environmental Toxins and Mental Health
NEUR 3402 [0.5]	Impact of Lifestyle and Social Interactions on Mental Health
NEUR 3403 [0.5]	Stress and Mental Health
NEUR 3501 [0.5]	Neurodegeneration and Aging

NEUR 3502 [0.5]	Neurodevelopmental Determinants of Mental Health	
4. 0.5 credit from:		0.5
NEUR 4001 [0.5]	Special Topics in Neuroscience	
NEUR 4002 [0.5]	Systematic Reviews and Meta-Analyses	
NEUR 4301 [0.5]	Neurobiology of Energy Homeostasis	
NEUR 4302 [0.5]	Sex and the Brain	
NEUR 4303 [0.5]	Indigenous Health & Mental Health	
NEUR 4305 [0.5]	Immune-Brain Interactions	
NEUR 4306 [0.5]	The Neural Basis of Addiction	
NEUR 4600 [0.5]	Advanced Lab in Neuroanatomy	
5. 0.5 credit from:		0.5
NEUR 4200 [0.5]	Seminar on Current Advances in Neuroscience	
NEUR 4202 [0.5]	Seminar on Current Research in Neuroscience and Psychiatric Conditions	
NEUR 4203 [0.5]	Seminar on Current Research in Neuroscience and Clinical Neurology	
6. 1.0 credit from:		1.0
NEUR 4904 [1.0]	Honours Research Thesis in Systematic Reviews or Meta-Analyses	
NEUR 4905 [1.0]	Honours Workshop	
NEUR 4906 [1.0]	Translational Approach to Indigenous Community Wellness	
NEUR 4907 [1.0]	Honours Essay	
NEUR 4908 [1.0]	Honours Research Thesis	
7. 0.5 credit in Advanced Science Faculty Electives		0.5
B. Credits Not Included in the Major CGPA (9.0 credits)		
8. 1.0 credits in:		1.0
CHEM 1001 [0.5]	General Chemistry I & General Chemistry II	
CHEM 1002 [0.5]		
9. 1.0 credit from:		1.0
COMP 1005 [0.5]	Introduction to Computer Science I & Introduction to Computer Science II	
COMP 1006 [0.5]		
or		
PHYS 1007 [0.5]	Elementary University Physics I & PHYS 1008 [0.5] Elementary University Physics II	
or		
PHYS 1107 [0.5]	Introductory University Physics I & PHYS 1108 [0.5] Introductory University Physics II	
or		
PSYC 1001 [0.5]	Introduction to Psychology I & PSYC 1002 [0.5] Introduction to Psychology II	
10. 0.5 credit in:		0.5
MATH 1107 [0.5]	Linear Algebra I	
MATH 1007 [0.5]	Elementary Calculus I (9. 0.5 credit from:)	
11. 1.0 credit in:		1.0
BIOL 2107 [0.5]	Fundamentals of Genetics	
BIOL 2201 [0.5]	Cell Biology and Biochemistry or BIOL 2200 [0.5] Cellular Biochemistry	
12. 1.0 credit in Science Continuation Courses		1.0

13. 2.0 credits in approved courses outside the faculties of Science and Engineering and Design (may include ISAP 1000)	2.0
14. 2.5 credits in free electives	2.5
Total Credits	20.0

Neuroscience and Mental Health

B.Sc. Major (20.0 credits)

A. Credits Included in the Major CGPA (11.0 credits)

1. 6.0 credits in:	6.0
NEUR 1202 [0.5] Neuroscience of Mental Health and Psychiatric Conditions	
NEUR 1203 [0.5] Neuroscience of Mental Health and Neurological Conditions	
NEUR 2001 [0.5] Introduction to Research Methods in Neuroscience	
NEUR 2002 [0.5] Introduction to Statistics in Neuroscience	
NEUR 2004 [0.5] Fundamentals of Scientific Writing in Neuroscience	
NEUR 2201 [0.5] Cellular and Molecular Neuroscience	
NEUR 2202 [0.5] Neurodevelopment and Plasticity	
NEUR 3001 [0.5] Data Analysis in Neuroscience I	
NEUR 3002 [0.5] Data Analysis in Neuroscience II	
NEUR 3204 [0.5] Neuropharmacology	
NEUR 3206 [0.5] Sensory and Motor Neuroscience	
NEUR 3207 [0.5] Systems Neuroscience	
2. 1.0 credit in:	1.0
BIOL 1103 [0.5] Foundations of Biology I	
BIOL 1104 [0.5] Foundations of Biology II	
3. 1.5 credits from:	1.5
NEUR 3003 [0.5] Epidemiology in Neuroscience	
NEUR 3301 [0.5] Genetics of Mental Health	
NEUR 3303 [0.5] The Neuroscience of Consciousness	
NEUR 3304 [0.5] Hormones and Behaviour	
NEUR 3401 [0.5] Environmental Toxins and Mental Health	
NEUR 3402 [0.5] Impact of Lifestyle and Social Interactions on Mental Health	
NEUR 3403 [0.5] Stress and Mental Health	
NEUR 3501 [0.5] Neurodegeneration and Aging	
NEUR 3502 [0.5] Neurodevelopmental Determinants of Mental Health	
4. 1.0 credit from:	1.0
NEUR 4001 [0.5] Special Topics in Neuroscience	
NEUR 4002 [0.5] Systematic Reviews and Meta-Analyses	
NEUR 4301 [0.5] Neurobiology of Energy Homeostasis	
NEUR 4302 [0.5] Sex and the Brain	
NEUR 4303 [0.5] Indigenous Health & Mental Health	
NEUR 4305 [0.5] Immune-Brain Interactions	
NEUR 4306 [0.5] The Neural Basis of Addiction	
NEUR 4600 [0.5] Advanced Lab in Neuroanatomy	
5. 1.0 credit from:	1.0
NEUR 4200 [0.5] Seminar on Current Advances in Neuroscience	

NEUR 4202 [0.5] Seminar on Current Research in Neuroscience and Psychiatric Conditions	
NEUR 4203 [0.5] Seminar on Current Research in Neuroscience and Clinical Neurology	
6. 0.5 credit in Advanced Science Faculty Electives	0.5
B. Credits Not Included in the Major CGPA (9.0 credits)	
7. 1.0 credits in:	1.0
CHEM 1001 [0.5] General Chemistry I & CHEM 1002 [0.5] General Chemistry II	
8. 1.0 credits in:	1.0
COMP 1005 [0.5] Introduction to Computer Science I & COMP 1006 [0.5] Introduction to Computer Science II	
or	
PHYS 1007 [0.5] Elementary University Physics I & PHYS 1008 [0.5] Elementary University Physics II	
or	
PHYS 1107 [0.5] Introductory University Physics I & PHYS 1108 [0.5] Introductory University Physics II	
or	
PSYC 1001 [0.5] Introduction to Psychology I & PSYC 1002 [0.5] Introduction to Psychology II	
9. 0.5 credit from:	0.5
MATH 1007 [0.5] Elementary Calculus I	
MATH 1107 [0.5] Linear Algebra I	
10. 1.0 credit in:	1.0
BIOL 2107 [0.5] Fundamentals of Genetics	
BIOL 2201 [0.5] Cell Biology and Biochemistry or BIOL 2200 [0.5] Cellular Biochemistry	
11. 1.0 credit in Science Continuation courses (not in NEUR)	1.0
12. 2.0 credits in approved courses outside the faculties of Science and Engineering and Design (may include ISAP 1000)	2.0
13. 2.5 credits in free electives	2.5
Total Credits	20.0

Neuroscience and Mental Health

B.Sc. (15.0 credits)

A. Credits Included in the Major CGPA (7.5 credits)

1. 5.0 credits in:	5.0
NEUR 1202 [0.5] Neuroscience of Mental Health and Psychiatric Conditions	
NEUR 1203 [0.5] Neuroscience of Mental Health and Neurological Conditions	
NEUR 2001 [0.5] Introduction to Research Methods in Neuroscience	
NEUR 2002 [0.5] Introduction to Statistics in Neuroscience	
NEUR 2004 [0.5] Fundamentals of Scientific Writing in Neuroscience	
NEUR 2201 [0.5] Cellular and Molecular Neuroscience	
NEUR 2202 [0.5] Neurodevelopment and Plasticity	
NEUR 3204 [0.5] Neuropharmacology	
NEUR 3206 [0.5] Sensory and Motor Neuroscience	
NEUR 3207 [0.5] Systems Neuroscience	

2. 1.0 credit in:	1.0
BIOL 1103 [0.5]	Foundations of Biology I
BIOL 1104 [0.5]	Foundations of Biology II
3. 1.5 credits from:	1.5
NEUR 3003 [0.5]	Epidemiology in Neuroscience
NEUR 3301 [0.5]	Genetics of Mental Health
NEUR 3303 [0.5]	The Neuroscience of Consciousness
NEUR 3304 [0.5]	Hormones and Behaviour
NEUR 3401 [0.5]	Environmental Toxins and Mental Health
NEUR 3402 [0.5]	Impact of Lifestyle and Social Interactions on Mental Health
NEUR 3403 [0.5]	Stress and Mental Health
NEUR 3501 [0.5]	Neurodegeneration and Aging
NEUR 3502 [0.5]	Neurodevelopmental Determinants of Mental Health
B. Credits Not Included in the Major CGPA (7.5 credits)	
4. 1.0 credit in:	1.0
CHEM 1001 [0.5]	General Chemistry I
& CHEM 1002 [0.5]	General Chemistry II
5. 1.0 credit in:	1.0
COMP 1005 [0.5]	Introduction to Computer Science I
& COMP 1006 [0.5]	Introduction to Computer Science II
or	
PHYS 1007 [0.5]	Elementary University Physics I
& PHYS 1008 [0.5]	Elementary University Physics II
or	
PHYS 1107 [0.5]	Introductory University Physics I
& PHYS 1108 [0.5]	Introductory University Physics II
or	
PSYC 1001 [0.5]	Introduction to Psychology I
& PSYC 1002 [0.5]	Introduction to Psychology II
6. 0.5 credit from:	0.5
MATH 1007 [0.5]	Elementary Calculus I
MATH 1107 [0.5]	Linear Algebra I
7. 1.0 credit in:	1.0
BIOL 2107 [0.5]	Fundamentals of Genetics
BIOL 2201 [0.5]	Cell Biology and Biochemistry
8. 1.0 credit in Science Continuation courses (not in NEUR)	1.0
9. 2.0 credits in approved courses outside the faculties of Science and Engineering and Design (may include ISAP 1000)	2.0
10. 1.0 credit in free electives	1.0
Total Credits	15.0

Neuroscience and Biology

B.Sc. Combined Honours (20.0 credits)

A. Credits Included in the Major CGPA (14.5 credits)

1. 5.5 credits in:	5.5
NEUR 1202 [0.5]	Neuroscience of Mental Health and Psychiatric Conditions
NEUR 1203 [0.5]	Neuroscience of Mental Health and Neurological Conditions
NEUR 2001 [0.5]	Introduction to Research Methods in Neuroscience

NEUR 2002 [0.5]	Introduction to Statistics in Neuroscience
NEUR 2201 [0.5]	Cellular and Molecular Neuroscience
NEUR 2202 [0.5]	Neurodevelopment and Plasticity
NEUR 3001 [0.5]	Data Analysis in Neuroscience I
NEUR 3002 [0.5]	Data Analysis in Neuroscience II
NEUR 3204 [0.5]	Neuropharmacology
NEUR 3206 [0.5]	Sensory and Motor Neuroscience
NEUR 3207 [0.5]	Systems Neuroscience
2. 3.0 credits in:	3.0
BIOL 1103 [0.5]	Foundations of Biology I
BIOL 1104 [0.5]	Foundations of Biology II
BIOL 2001 [0.5]	Animals: Form and Function
BIOL 2104 [0.5]	Introductory Genetics
BIOL 2200 [0.5]	Cellular Biochemistry
BIOL 3305 [0.5]	Human and Comparative Physiology
3. 1.5 credits in BIOL or BIOC at the 3000 level or above	1.5
4. 1.0 credit from:	1.0
NEUR 3301 [0.5]	Genetics of Mental Health
NEUR 3303 [0.5]	The Neuroscience of Consciousness
NEUR 3304 [0.5]	Hormones and Behaviour
NEUR 3401 [0.5]	Environmental Toxins and Mental Health
NEUR 3402 [0.5]	Impact of Lifestyle and Social Interactions on Mental Health
NEUR 3403 [0.5]	Stress and Mental Health
NEUR 3501 [0.5]	Neurodegeneration and Aging
NEUR 3502 [0.5]	Neurodevelopmental Determinants of Mental Health
NEUR 4301 [0.5]	Neurobiology of Energy Homeostasis
NEUR 4302 [0.5]	Sex and the Brain
NEUR 4303 [0.5]	Indigenous Health & Mental Health
NEUR 4305 [0.5]	Immune-Brain Interactions
NEUR 4306 [0.5]	The Neural Basis of Addiction
NEUR 4600 [0.5]	Advanced Lab in Neuroanatomy
5. 2.0 credits from:	2.0
BIOC 4007 [0.5]	Membrane Biochemistry
BIOL 2600 [0.5]	Ecology
BIOL 2301 [0.5]	Biotechnology I
BIOL 2303 [0.5]	Microbiology
BIOL 3307 [0.5]	Advanced Human Anatomy and Physiology
BIOL 3605 [0.5]	Field Course I
BIOL 3609 [0.5]	Evolutionary Concepts
BIOL 3802 [0.5]	Animal Behaviour
BIOL 3804 [0.5]	Social Evolution
BIOL 4306 [0.5]	Animal Neurophysiology
BIOL 4317 [0.5]	Neuroethology: The Neural Basis of Animal Behaviour
BIOL 4802 [0.5]	Advanced Animal Behaviour
CHEM 2204 [0.5]	Organic Chemistry II
6. 0.5 credit from:	0.5
NEUR 4200 [0.5]	Seminar on Current Advances in Neuroscience

NEUR 4202 [0.5]	Seminar on Current Research in Neuroscience and Psychiatric Conditions	
NEUR 4203 [0.5]	Seminar on Current Research in Neuroscience and Clinical Neurology	
7. 1.0 credit from:		1.0
NEUR 4905 [1.0]	Honours Workshop	
NEUR 4907 [1.0]	Honours Essay	
NEUR 4908 [1.0]	Honours Research Thesis	
BIOL 4905 [1.0]	Honours Workshop	
BIOL 4907 [1.0]	Honours Essay and Research Proposal	
BIOL 4908 [1.0]	Honours Research Thesis	
B. Credits not included in the Major CGPA (5.5 credits)		
8. 1.0 credit in:		1.0
MATH 1007 [0.5]	Elementary Calculus I	
MATH 1107 [0.5]	Linear Algebra I	
9. 1.5 credits in:		1.5
CHEM 1001 [0.5]	General Chemistry I	
& CHEM 1002 [0.5]	General Chemistry II	
CHEM 2203 [0.5]	Organic Chemistry I	
10. 1.0 credit in:		1.0
PHYS 1007 [0.5]	Elementary University Physics I	
& PHYS 1008 [0.5]	Elementary University Physics II	
11. 2.0 credits in approved courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000)		2.0
Total Credits		20.0

Minor in Neuroscience and Mental Health (4.0 credits)

The Minor in Neuroscience is available to students registered in degree programs other than those offered by the Department of Neuroscience or the Bachelor of Science in Nursing with RN Prescribing with Concentration in Neuroscience and Mental Health.

Students are required to present a Minor CGPA of 4.00 or higher at graduation in order to be awarded a Minor in Neuroscience and Mental Health.

Requirements:

1. 2.0 credits in:		2.0
NEUR 1202 [0.5]	Neuroscience of Mental Health and Psychiatric Conditions	
NEUR 1203 [0.5]	Neuroscience of Mental Health and Neurological Conditions	
NEUR 2201 [0.5]	Cellular and Molecular Neuroscience	
NEUR 2202 [0.5]	Neurodevelopment and Plasticity	
2. 2.0 credits from:		2.0
NEUR 3204 [0.5]	Neuropharmacology	
NEUR 3301 [0.5]	Genetics of Mental Health	
NEUR 3303 [0.5]	The Neuroscience of Consciousness	
NEUR 3304 [0.5]	Hormones and Behaviour	
NEUR 3401 [0.5]	Environmental Toxins and Mental Health	

NEUR 3402 [0.5]	Impact of Lifestyle and Social Interactions on Mental Health	
NEUR 3403 [0.5]	Stress and Mental Health	
NEUR 3501 [0.5]	Neurodegeneration and Aging	
NEUR 3502 [0.5]	Neurodevelopmental Determinants of Mental Health	
NEUR 4301 [0.5]	Neurobiology of Energy Homeostasis	
NEUR 4302 [0.5]	Sex and the Brain	
NEUR 4303 [0.5]	Indigenous Health & Mental Health	
NEUR 4306 [0.5]	The Neural Basis of Addiction	
Total Credits		4.0

Students enrolled in the Neuroscience and Mental Health programs should consult with the Department of Neuroscience when planning their program or selecting courses. Those enrolled in the Neuroscience Combined Honours program should consult with either the Department of Biology or the Department of Neuroscience.

B.Sc. Regulations

The regulations presented in this section apply to all Bachelor of Science programs. In addition to the requirements presented here, students must satisfy the University regulations common to all undergraduate students including the process of Academic Continuation Evaluation (see the *Academic Regulations of the University* section of this Calendar).

Breadth Requirement for the B.Sc.

Students in a Bachelor of Science program must present the following credits at graduation:

- 2.0 credits in Science Continuation courses not in the major discipline; **students completing a double major are considered to have completed this requirement providing they have 2.0 credits in Science Continuation courses in each of the two majors;**
- 2.0 credits in courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000)

In most cases, the requirements for individual B.Sc. programs, as stated in this Calendar, contain these requirements, explicitly or implicitly.

Students admitted to B.Sc. programs by transfer from another institution must present at graduation (whether taken at Carleton or elsewhere):

- 2.0 credits in courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000) if the student received fewer than 10.0 transfer credits; or,
- 1.0 credit in courses outside of the faculties of Science and Engineering and Design (may include ISAP 1000) if the student received 10.0 or more transfer credits.

Declared and Undeclared Students

Degree students are considered "Undeclared" if they have been admitted to a degree, but have not yet selected and been accepted into a program within that degree. The status "Undeclared" is available only in the B.A. and

B.Sc. degrees. Undeclared students must apply to enter a program upon or before completing 3.5 credits.

Change of Program within the B.Sc. Degree

To transfer to a program within the B.Sc. degree, applicants must normally be *Eligible to Continue* (EC) in the new program, by meeting the CGPA thresholds described in Section 3.1.9 of the *Academic Regulations of the University*.

Applications to declare or change programs within the B.Sc. degree must be made online through Carleton Central by completing a Change of Program Elements (COPE) application form within the published deadlines. Acceptance into a program, or into a program element or option, is subject to any enrolment limitations, and/or specific program, program element or option requirements as published in the relevant Calendar entry.

Minors, Concentrations, and Specializations

Students may add a Minor, Concentration, or Specialization by completing a Change of Program Elements (COPE) application form online through Carleton Central. Acceptance into a Minor, Concentration, or Specialization normally requires that the student be *Eligible to Continue* (EC) and is meeting the minimum CGPAs described in Section 3.1.9 of the *Academic Regulations of the University*, as well as being subject to any specific requirements of the intended Minor, Concentration, or Specialization as published in the relevant Calendar entry.

Experimental Science Requirement

Students in a B.Sc. degree program must present at graduation at least two full credits of Experimental Science chosen from two different departments or institutes from the list below:

Approved Experimental Science Courses

Biochemistry

BIOC 2200 [0.5]	Cellular Biochemistry
BIOC 3103 [0.5]	Experimental Biochemistry I: Principles and Practices
BIOC 3104 [0.5]	Experimental Biochemistry II: Research Experience
BIOC 4001 [0.5]	Methods in Biochemistry
BIOC 4201 [0.5]	Advanced Cell Culture and Tissue Engineering

Biology

BIOL 1103 [0.5]	Foundations of Biology I
BIOL 1104 [0.5]	Foundations of Biology II
BIOL 2001 [0.5]	Animals: Form and Function
BIOL 2002 [0.5]	Plants: Form and Function
BIOL 2104 [0.5]	Introductory Genetics
BIOL 2200 [0.5]	Cellular Biochemistry
BIOL 2600 [0.5]	Ecology

Chemistry

CHEM 1001 [0.5]	General Chemistry I
CHEM 1002 [0.5]	General Chemistry II
CHEM 2103 [0.5]	Physical Chemistry I
CHEM 2203 [0.5]	Organic Chemistry I
CHEM 2204 [0.5]	Organic Chemistry II

CHEM 2302 [0.5]	Analytical Chemistry I
CHEM 2303 [0.5]	Analytical Chemistry II
CHEM 2800 [0.5]	Foundations for Environmental Chemistry

Earth Sciences

ERTH 1002 [0.5]	The Earth and Life Odyssey: A Journey Through Billions of Years
ERTH 2102 [0.5]	Mineralogy to Petrology
ERTH 2404 [0.5]	Engineering Geoscience
ERTH 2802 [0.5]	Field Geology I
ERTH 3111 [0.5]	Vertebrate Evolution: Mammals, Reptiles, and Birds
ERTH 3112 [0.5]	Vertebrate Evolution: Fish and Amphibians
ERTH 3204 [0.5]	Mineral Deposits
ERTH 3205 [0.5]	Physical Hydrogeology

Food Sciences

FOOD 3001 [0.5]	Food Chemistry
FOOD 3002 [0.5]	Food Analysis
FOOD 3005 [0.5]	Food Microbiology

Geography

GEOG 1010 [0.5]	Global Environmental Systems
GEOG 3108 [0.5]	Soil Properties

Neuroscience

NEUR 3206 [0.5]	Sensory and Motor Neuroscience
NEUR 3207 [0.5]	Systems Neuroscience
NEUR 4600 [0.5]	Advanced Lab in Neuroanatomy

Physics

PHYS 1001 [0.5]	Foundations of Physics I
PHYS 1002 [0.5]	Foundations of Physics II
PHYS 1003 [0.5]	Introductory Mechanics and Thermodynamics
PHYS 1004 [0.5]	Introductory Electromagnetism and Wave Motion
PHYS 1007 [0.5]	Elementary University Physics I
PHYS 1008 [0.5]	Elementary University Physics II
PHYS 2007 [0.5]	Second Year Physics Laboratory: Selected Experiments and Seminars
PHYS 3007 [0.5]	Third Year Physics Laboratory: Selected Experiments and Seminars
PHYS 3606 [0.5]	Modern Physics II
PHYS 3608 [0.5]	Modern Applied Physics

Course Categories for B.Sc. Programs

Science Geography Courses

GEOG 1010 [0.5]	Global Environmental Systems
GEOG 2006 [0.5]	Introduction to Quantitative Research
GEOG 2013 [0.5]	Weather and Water
GEOG 2014 [0.5]	The Earth's Surface
GEOG 3003 [0.5]	Quantitative Geography
GEOG 3010 [0.5]	Field Methods in Physical Geography
GEOG 3102 [0.5]	Geomorphology
GEOG 3103 [0.5]	Watershed Hydrology
GEOG 3104 [0.5]	Principles of Biogeography

GEOG 3105 [0.5]	Climate and Atmospheric Change
GEOG 3106 [0.5]	Aquatic Science and Management
GEOG 3108 [0.5]	Soil Properties
GEOG 4000 [0.5]	Field Studies
GEOG 4005 [0.5]	Directed Studies in Geography
GEOG 4013 [0.5]	Cold Region Hydrology
GEOG 4017 [0.5]	Global Biogeochemical Cycles
GEOG 4101 [0.5]	Two Million Years of Environmental Change
GEOG 4103 [0.5]	Water Resources Engineering
GEOG 4104 [0.5]	Microclimatology
GEOG 4108 [0.5]	Permafrost

Science Psychology Courses

PSYC 2001 [0.5]	Introduction to Research Methods in Psychology
PSYC 2002 [0.5]	Introduction to Statistics in Psychology
PSYC 2700 [0.5]	Introduction to Cognitive Psychology
PSYC 3000 [1.0]	Design and Analysis in Psychological Research
PSYC 3506 [0.5]	Cognitive Development
PSYC 3700 [1.0]	Cognition (Honours Seminar)
PSYC 3702 [0.5]	Perception
PSYC 2307 [0.5]	Human Neuropsychology I
PSYC 3307 [0.5]	Human Neuropsychology II

Science Continuation Courses

A course at the 2000 level or above may be used as a Science Continuation credit in a B.Sc. program if it is not in the student's major discipline, and is chosen from the following:

BIOC (Biochemistry)

BIOL (Biology) except BIOL 4810 which may be used only as a free elective for any B.Sc. program. Biochemistry students may use BIOL 2005 only as a free elective.

CHEM (Chemistry)

COMP (Computer Science) A maximum of two half-credits at the 1000-level in COMP, excluding COMP 1001 may be used as Science Continuation credits.

ERTH (Earth Sciences), except ERTH 2415 which may be used only as a free elective for any B.Sc. program. Students in Earth Sciences programs may use ERTH 2401, ERTH 2402, and ERTH 2403 only as free electives.

Engineering. Students wishing to register in Engineering courses must obtain the permission of the Faculty of Engineering and Design.

ENSC (Environmental Science)

FOOD (Food Science and Nutrition)

GEOM (Geomatics)

HLTH (Health Sciences)

ISAP (Interdisciplinary Science Practice)

MATH (Mathematics)

NEUR (Neuroscience)

PHYS (Physics), except PHYS 2903

Science Geography Courses (see list above)

Science Psychology Courses (see list above)

STAT (Statistics)

TSES (Technology, Society, Environment) except TSES 2305. Biology students may use these courses only as free electives. Integrated Science and Environmental Science students may include these courses in their programs but may not count them as part of the Science Sequence.

Science Faculty Electives

Science Faculty Electives are courses at the 1000-4000 level chosen from:

BIOC (Biochemistry) except BIOC 1900 which may be used only as a free elective for any B.Sc. program.

BIOL (Biology) except BIOL 4810 which may be used only as a free elective for any B.Sc. program. Biology & Biochemistry students may use BIOL 1010 and BIOL 2005 only as free electives.

CHEM (Chemistry) except CHEM 1003, CHEM 1004 and CHEM 1007

COMP (Computer Science) except COMP 1001

ERTH (Earth Sciences) except ERTH 1004 and ERTH 2415. Earth Sciences students may use ERTH 2401, ERTH 2402 and ERTH 2403 only as free electives.

Engineering

ENSC (Environmental Science)

FOOD (Food Science and Nutrition)

GEOM (Geomatics)

HLTH (Health Science)

ISAP (Interdisciplinary Science Practice)

MATH (Mathematics)

NEUR (Neuroscience)

PHYS (Physics) except PHYS 1901, PHYS 1902, PHYS 1905, and PHYS 2903

Science Geography (see list above)

Science Psychology (see list above)

STAT (Statistics)

TSES (Technology, Society, Environment) Biology students may use these courses only as free electives.

Advanced Science Faculty Electives

Advanced Science Faculty Electives are courses at the 2000-4000 level chosen from the Science Faculty Electives list above.

Approved Courses Outside the Faculties of Science and Engineering and Design (may include ISAP 1000)

All courses offered by the Faculty of Arts and Social Sciences, the Faculty of Public and Global Affairs, and the Sprott School of Business are approved as Arts or Social Sciences courses EXCEPT FOR: All Science Geography courses (see list above), all Geomatics (GEOM) courses, all Science Psychology courses (see list above). ISAP 1000 may be used as an Approved Course Outside the Faculties of Science and Engineering and Design.

Free Electives

Any course is allowable as a Free Elective providing it is not prohibited (see below). Students are expected to comply with prerequisite requirements and enrolment restrictions for all courses as published in this Calendar.

Courses Allowable Only as Free Electives in any B.Sc. Program

BIOL 4810 [0.5]	Education Research in Undergraduate Science
CHEM 1003 [0.5]	The Chemistry of Food, Health and Drugs
CHEM 1004 [0.5]	Drugs and the Human Body
CHEM 1007 [0.5]	Chemistry of Art and Artifacts
ERTH 1004 [0.5]	Earth's Epic Tale: A Story Across Billions of Years
ERTH 2415 [0.5]	Natural Disasters
ISCI 1001 [0.5]	Introduction to the Environment
ISCI 2000 [0.5]	Natural Laws
ISCI 2002 [0.5]	Human Impacts on the Environment
PHYS 1901 [0.5]	Planetary Astronomy
PHYS 1902 [0.5]	From our Star to the Cosmos
PHYS 1905 [0.5]	Physics Behind Everyday Life
PHYS 2903 [0.5]	Physics Towards the Future

Prohibited Courses

The following courses are not acceptable for credit in any B.Sc. program:

COMP 1001 [0.5]	Introduction to Computational Thinking for Arts and Social Science Students
MATH 1009 [0.5]	Mathematics for Business
MATH 1119 [0.5]	Linear Algebra: with Applications to Business
MATH 1401 [0.5]	Elementary Mathematics for Economics I
MATH 1402 [0.5]	Elementary Mathematics for Economics II
all 0000-level courses	

Co-operative Education

For more information about how to apply for the Co-op program and how the Co-op program works please visit the Co-op website.

All students participating in the Co-op program are governed by the Undergraduate Co-operative Education Policy.

Undergraduate Co-operative Education Policy

Admission Requirements

Students can apply to Co-op in one of two ways: directly from high school, or after beginning a degree program at Carleton.

If a student applies to a degree program with a Co-op option from high school, their university grades will be reviewed two terms to one year prior to their first work term to ensure they meet the academic requirements after their first or second year of study. The time at which the evaluation takes place depends on the program of study. Students will automatically receive an admission decision via their Carleton email account.

Students who did not request Co-op at the time they applied to Carleton can request Co-op after they begin their university studies. To view application instructions and deadlines, please visit carleton.ca/co-op.

To be admitted to Co-op, a student must successfully complete 5.0 or more credits that count towards their

degree, meet the minimum CGPA requirement(s) for the student's Co-op option, and fulfil any specified course prerequisites. To see the unique admission and continuation requirements for each Co-op option, please refer to the specific degree programs listed in the Undergraduate Calendar.

Participation Requirements

Co-op Participation Agreement

All students must adhere to the policies found within the Co-op Participation Agreement.

COOP 1000

Once a student has been admitted to the Co-op Program, they will be given access to register in COOP 1000. This zero-credit online course must be completed at least two terms prior to the student's first work term.

Communication with the Co-op Office

Students must maintain contact with the Co-op Office during their job search and while on a work term. All email communication will be conducted via the students' Carleton email account.

Employment

Although every effort is made to ensure a sufficient number of job postings for all Co-op students, no guarantee of employment can be made. The Co-op job search process is competitive, and success is dependent upon factors such as current market conditions, academic performance, skills, motivation, and level of commitment to the job search. It is the student's responsibility to apply for positions via the Co-op job board in addition to actively conducting a self-directed job search. Students who do not obtain a co-op work term are expected to continue with their academic studies. It should be noted that hiring priority for positions within the Federal Government of Canada is given to Canadian citizens.

Registration

- Students must be registered as full-time during all fall and winter study terms beginning the term in which they enroll in COOP 1000.
- Students will be registered in a Co-op Work Term course while at work. This course does not carry academic course credit, but is noted on academic transcripts.
- Students may register in a 0.5 credit during a work term, provided the course is offered during the evening or is offered asynchronously online.
- Students must have at least one term of full-time studies left to complete following their final co-op work term. Students cannot end their degree on a work term.

Work Term Assessment and Evaluation

Work Term Evaluation

Employers are responsible for submitting to Carleton University final performance evaluations for their Co-op students at the end of their work terms.

Work Term Assessment

In order to successfully complete the co-op work term, students must receive a Satisfactory (SAT) grade on their Co-op Work Term Report, which they must submit at the completion of each four-month work term.

Graduation with the Co-op Designation

In order to graduate with the Co-op Designation, students must satisfy all requirements of the degree program in addition to the successful completion of three or four work terms (the number is dependent upon the student's academic program). Students found in violation of the Co-op Participation Agreement may have the Co-op Designation withheld.

Note: Participation in the co-op option will add up to one additional year for a student to complete their degree program.

Voluntary Withdrawal from the Co-op Option

Students who are currently on a co-op work term or who have already committed to a co-op work term either verbally or in writing may not leave the position and/or withdraw from the co-op option until they have completed the work term and all related requirements.

Involuntary or Required Withdrawal from the Co-op Option

Students may be removed from the Co-op Program for any of the following reasons:

1. Failure to achieve a grade of SAT in COOP 1000;
2. Failure to attend all interviews for positions to which the student has applied;
3. Declining more than one job offer during the job search;
4. Reneging on a co-op position that the student has accepted either verbally or in writing;
5. Continuing a job search after accepting a co-op position;
6. Dismissal from a work term by the co-op employer;
7. Leaving a work term without approval from the Co-op Management Team;
8. Receipt of an unsatisfactory work term evaluation;
9. Receiving a grade of UNS on the work term report.

International Students

All international students are required to possess a Co-op Work Permit issued by Immigration, Refugees and Citizenship Canada before they can begin working. The Co-operative Education Office will provide students with a letter of support to accompany their Co-op Work Permit application. Students are advised to discuss the application process and application requirements with the International Student Services Office.

Co-op Fees

All participating Co-op students are required to pay Co-op fees. For full details, please see the Co-op website.

B.Sc. Honours Neuroscience and Mental Health; B.Sc. Combined Honours Neuroscience and Biology: Co-op Admission and Continuation Requirements

- Maintain full-time status in each study term;
- Be eligible to work in Canada (for off-campus work);
- Have successfully completed COOP 1000 .

In addition to the following:

1. Registered as a full-time student in the B.Sc. Honours Neuroscience and Mental Health or B.Sc. Combined Honours Neuroscience and Biology program;
2. Successfully completed 5.0 or more credits;
3. Obtained an Overall CGPA of at least 6.50 and a Major CGPA of at least 8.00. These CGPAs must be maintained throughout the duration of the degree.

B.Sc. Honours Neuroscience and Mental Health and B.Sc. Combined Honours Neuroscience and Biology students must successfully complete three (3) work terms to obtain the Co-op Designation.

Work Term Course for Neuroscience and Mental Health: NEUR 3999

Work Term Course for Combined Honours Neuroscience and Biology: NEUR 3999, BIOL 3999

Work-Study Pattern:

Year 1		Year 2		Year 3		Year 4		Year 5	
Term	Pattern	Term	Pattern	Term	Pattern	Term	Pattern	Term	Pattern
Fall	S	Fall	S	Fall	S	Fall	W	Fall	S
Winter	S	Winter	S	Winter	S	Winter	W	Winter	S
Summer		Summer	W	Summer	W	Summer	W		

Legend

S: Study

W: Work

Admissions Information

Admission Requirements are for the 2026-27 year only, and are based on the Ontario High School System. Holding the minimum admission requirements only establishes eligibility for consideration. The cut-off averages for admission may be considerably higher than the minimum. See also the **General Admission and Procedures** section of this Calendar. An overall average of at least 70% is normally required to be considered for admission. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. The overall average required for admission is determined each year on a program by program basis. Consult admissions.carleton.ca for further details.

Note: Courses listed as *recommended* are not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Admissions Information

Admission requirements are based on the Ontario High School System. Prospective students can view the admission requirements through the Admissions website at admissions.carleton.ca. The overall average required for admission is determined each year on a program-by-program basis. Holding the minimum admission requirements only establishes eligibility for consideration; higher averages are required for admission to programs for which the demand for places by qualified applicants exceeds the number of places available. All programs have limited enrolment and admission is not guaranteed. Some programs may also require specific course prerequisites and prerequisite averages and/or supplementary admission portfolios. Consult admissions.carleton.ca for further details.

Note: If a course is listed as recommended, it is not mandatory for admission. Students who do not follow the recommendations will not be disadvantaged in the admission process.

Degrees

- B.Sc. (Honours)
- B.Sc. (Major)
- B.Sc.

Admission Requirements

B. Sc. Honours

First Year

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. For most programs including Biochemistry, Bioinformatics, Biotechnology, Chemistry, Combined Honours in Biology and Physics, Chemistry and Physics, Computational Biochemistry, Food Science, Nanoscience, Neuroscience and Biology, Neuroscience and Mental Health, and Psychology, the six 4U or M courses must include Advanced Functions, and two of Biology, Chemistry, Earth and Space Sciences, or Physics. (Calculus and Vectors is strongly recommended).

Specific Honours Admission Requirements

For the Honours programs in Earth Sciences, Environmental Science, Geomatics, Integrated Science, and Physical Geography, Calculus and Vectors may be substituted for Advanced Functions.

For the Honours programs in Physics and Applied Physics, and for double Honours in Mathematics and Physics, Calculus and Vectors is required in addition to Advanced Functions and one of 4U Physics, Chemistry, Biology, or Earth and Space Sciences. For all programs in Physics, 4U Physics is strongly recommended.

For Honours in Psychology, a 4U course in English is recommended.

For Honours in Environmental Science, a 4U course in Biology and Chemistry is recommended.

Advanced Standing

Applications for admission beyond first year will be assessed on their merits. Applicants must normally be

Eligible to Continue in their year level, in addition to meeting the CGPA thresholds described in Section 3.1.9 of the Academic Regulations of the University. Advanced standing will be granted only for those subjects deemed appropriate for the program and stream selected.

B.Sc. Major and B.Sc.

First Year

The Ontario Secondary School Diploma (OSSD) or equivalent including a minimum of six 4U or M courses. The six 4U or M courses must include Advanced Functions and two of Calculus and Vectors, Biology, Chemistry, Earth and Space Science, or Physics (Calculus and Vectors is strongly recommended). For the B.Sc. Major in Physics, 4U Physics is strongly recommended.

Advanced Standing

Applications for admission beyond first year will be assessed on their merits. Applicants must normally be *Eligible to Continue* (EC) in their year level. Advanced standing will be granted only for those subjects deemed appropriate for the program and stream selected.

Co-op Option

Direct Admission to the First Year of the Co-op Option
Applicants must:

1. meet the required overall admission cut-off average and prerequisite course average. These averages may be higher than the stated minimum requirements;
2. be registered as a full-time student in the Bachelor of Science Honours program;
3. be eligible to work in Canada (for off-campus work placements).

Note that meeting the above requirements only establishes eligibility for admission to the program. The prevailing job market may limit enrolment in the co-op option.

Note: continuation requirements for students previously admitted to the co-op option and admission requirements for the co-op option after beginning the program are described in the Co-operative Education Regulations section of this Calendar.

Neuroscience (NEUR) Courses

NEUR 1202 [0.5 credit]

Neuroscience of Mental Health and Psychiatric Conditions

An overview of mental health conditions, exploring biological, developmental, experiential and environmental factors that contribute to their onset and expression. Topics may include depressive and anxiety disorders, schizophrenia, autism, ADHD, eating disorders, sleep-related conditions, and substance use disorders. Precludes additional credit for NEUR 1201 (no longer offered).

Lecture three hours a week.

NEUR 1203 [0.5 credit]**Neuroscience of Mental Health and Neurological Conditions**

An overview of neurological conditions, examining biological, developmental, experiential and environmental contributions. Topics may include stroke, multiple sclerosis, seizure conditions, Parkinson's disease, ALS, chronic pain, Alzheimer's disease and concussion. Lectures three hours a week.

NEUR 2001 [0.5 credit]**Introduction to Research Methods in Neuroscience**

A general introduction to research process within neuroscience. Topics covered include research strategies, methods, and techniques; basic descriptive statistics; research communication; and responsible scientific conduct.

Precludes additional credit for PSYC 2000 and PSYC 2001.

Prerequisite(s): second-year standing.

Lecture three hours a week.

NEUR 2002 [0.5 credit]**Introduction to Statistics in Neuroscience**

A general introduction to statistical techniques employed within contemporary neuroscience. Topics covered include basic data analysis using descriptive and inferential statistics (t-tests, ANOVA, correlation, chi-square).

Precludes additional credit for ENST 2006, GEOG 2006, PSYC 2002.

Prerequisite(s): PSYC 2001 or NEUR 2001.

Lectures three hours a week, online labs/tutorials.

NEUR 2003 [0.5 credit]**Introduction to Techniques in Neuroscience**

Introduction to common techniques used in neuroscience research. Brain imaging, animal behaviour, electrophysiology, immunohistochemistry and microscopy, genomics, transgenics, cell culture, and DSM-IV-based clinical assessment.

Prerequisite(s): one of PSYC 1001, NEUR 1201, NEUR 1202 or NEUR 1203.

Lectures three hours a week.

NEUR 2004 [0.5 credit]**Fundamentals of Scientific Writing in Neuroscience**

Introduction to various forms of scientific writing appropriate to neuroscience, with a focus in fundamental skills in scientific writing.

Includes: Experiential Learning Activity

Prerequisite(s): second-year standing in a Neuroscience program and one of NEUR 1201, NEUR 1202 or NEUR 1203.

Lectures and workshops three hours a week.

NEUR 2201 [0.5 credit]**Cellular and Molecular Neuroscience**

Core principles in cellular and molecular neuroscience, including signal transmission along and between neurons, ion channels and transporters, intracellular signaling pathways, and regulation of gene expression.

Precludes additional credit for PSYC 3200 (no longer offered) and NEUR 3200 (no longer offered).

Prerequisite(s): Either NEUR 1201 and NEUR 1203, or NEUR 1202 and NEUR 1203, or both BIOL 1103 and BIOL 1104.

Lectures three hours a week, online labs.

NEUR 2202 [0.5 credit]**Neurodevelopment and Plasticity**

Core principles in nervous system development from embryogenesis to plasticity in the adult brain. Topics include neural induction, neurogenesis, apoptosis, neuronal migration and axon growth, synaptogenesis and synaptic pruning both under normal conditions and in psychopathology.

Precludes additional credit for PSYC 3200 (no longer offered) and NEUR 3200 (no longer offered).

Prerequisite(s): NEUR 1202 and NEUR 1203.

Lectures three hours a week, online labs.

NEUR 2801 [0.5 credit]**Neuroscience and Creativity**

Altered brain function related to mental health or substance use has often inspired cultural works including films, music, visual art and literature. This course explores the neurobiological basis of creativity in individuals with and without mental health conditions.

Prerequisite(s): one of PSYC 1001, NEUR 1201, NEUR 1202 or NEUR 1203.

Lectures and seminars three hours a week.

NEUR 3001 [0.5 credit]**Data Analysis in Neuroscience I**

Introducing various software for analyzing neuroscience data. Dealing with real data, drawing graphs, application of descriptive and inferential statistics through the general linear model, assumptions of parametric tests, robust statistics, confidence intervals, correlations, use of appropriate statistical methods and interpretation of results.

Includes: Experiential Learning Activity

Prerequisite(s): PSYC 2001 and PSYC 2002, or NEUR 2001 and NEUR 2002.

Lectures three hours a week, online labs/workshops.

NEUR 3002 [0.5 credit]**Data Analysis in Neuroscience II**

Use of software for analyzing neuroscience data. Statistical techniques typically include nonparametric tests, t tests, and various forms of both ANOVA and regression including robust statistical tests, with a focus on the practical application of appropriate statistical methods and interpretation of results.

Includes: Experiential Learning Activity

Prerequisite(s): NEUR 3001.

Lectures three hours a week, online labs/workshops.

NEUR 3003 [0.5 credit]**Epidemiology in Neuroscience**

Introduction to the principles and methods of epidemiology, study designs, measures of effect, sources of error, confounding, bias, internal and external validity, and causality. The course also will provide an overview of the epidemiological features, and risk factors for common neurological disorders.

Precludes additional credit for HLTH 3201.

Prerequisite(s): NEUR 2002.

Lectures three hours a week.

NEUR 3203 [0.5 credit]**Field Course in Animal Behaviour**

Offered in the Department of Biology as BIOL 3605. Only those modules dealing with animal behaviour topics may be offered for Neuroscience credit.

Includes: Experiential Learning Activity

Also listed as BIOL 3605.

Precludes additional credit for PSYC 3203.

Prerequisite(s): permission of the department.

NEUR 3204 [0.5 credit]**Neuropharmacology**

Overview of chemical neurotransmission and key neurotransmitter systems. A description of licit and illicit drugs covering topics that range from historical perspectives to pharmacology to mechanisms of action in the brain. Discussion of neurochemical basis of psychiatric diseases including anxiety, depression and schizophrenia. Precludes additional credit for PSYC 3204 (no longer offered).

Prerequisite(s): NEUR 2200 or NEUR 2201.

Lectures and seminars three hours a week.

NEUR 3206 [0.5 credit]**Sensory and Motor Neuroscience**

Exploration of major topics in sensory processing and motor control, with a focus on underlying mechanisms and neurobiological principles. Topics include all sensory systems (such as vision, somatosensation and audition) plus motor system components including lower and upper motor neurons, basal ganglia, and cerebellum.

Includes: Experiential Learning Activity

Precludes additional credit for PSYC 3200 (no longer offered), NEUR 3200 (no longer offered), PSYC 3202 (no longer offered) and NEUR 3202 (no longer offered).

Prerequisite(s): NEUR 1201 or both NEUR 1202 and NEUR 1203, and either NEUR 2200 or both NEUR 2201 and NEUR 2202.

Lectures three hours a week, laboratory four hours a week.

NEUR 3207 [0.5 credit]**Systems Neuroscience**

Neural systems underlying complex behaviours including emotion, motivation, and sleep, and the role of association cortices in brain function.

Includes: Experiential Learning Activity

Precludes additional credit for NEUR 3200 (no longer offered) and PSYC 3200 (no longer offered).

Prerequisite(s): NEUR 3206.

Lectures three hours a week, laboratory four hours a week.

NEUR 3301 [0.5 credit]**Genetics of Mental Health**

Many common mental health conditions have a genetic component. This course explores how genetic susceptibility is identified and examines the genetic, genomic and epigenetic mechanisms that may influence vulnerability to these conditions.

Prerequisite(s): BIOL 2104 or BIOL 2107, and NEUR 2200 or NEUR 2201.

Lectures three hours a week.

NEUR 3303 [0.5 credit]**The Neuroscience of Consciousness**

Consciousness remains one of the least understood aspects of the nervous system. This course explores neural mechanisms underlying consciousness, changes in consciousness associated with sleep, coma, vegetative states, drugs, and other stimuli, and considers the evolutionary basis of consciousness, and its relationship with awareness.

Prerequisite(s): NEUR 2200 or NEUR 2202.

Lectures three hours a week.

NEUR 3304 [0.5 credit]**Hormones and Behaviour**

The effects of hormones throughout life at all levels of the nervous system. The role of hormones in mediating behaviours that are both basic (feeding, reproduction and social interactions) and complex (motivation, emotion, learning and memory).

Prerequisite(s): NEUR 2200 or both NEUR 2201 and NEUR 2202.

Lectures three hours a week.

NEUR 3401 [0.5 credit]**Environmental Toxins and Mental Health**

Exposure to environmental toxins from the air, water or food can interfere with neuronal function, alter neurodevelopment, and damage the brain. This course will explore associations between toxins and diseases such as Parkinson's disease, multiple sclerosis and depression, focusing on mechanisms underlying development of pathology.

Prerequisite(s): NEUR 2200 or both NEUR 2201 and NEUR 2202.

Lectures three hours a week.

NEUR 3402 [0.5 credit]**Impact of Lifestyle and Social Interactions on Mental Health**

Healthy lifestyle choices and positive social interactions can reduce the incidence of pathological conditions such as depression, obesity, cardiovascular disease and impaired immunity. This course focuses on psychosocial and neurobiological mechanisms that underlie the relationship between lifestyle, social interactions and health.

Prerequisite(s): NEUR 2200 or both NEUR 2201 and NEUR 2202.

Lectures three hours a week.

NEUR 3403 [0.5 credit]**Stress and Mental Health**

Stressful events can have profound repercussions on physical and psychological well-being. This course examines the psychosocial and biological processes by which stressors predispose to both physical (immune-related disorders, diabetes, heart disease) and psychological (acute stress disorder, posttraumatic stress disorder, depression, anxiety) pathologies.

Prerequisite(s): NEUR 2200 or both NEUR 2201 and NEUR 2202.

Lectures three hours a week.

NEUR 3501 [0.5 credit]**Neurodegeneration and Aging**

Perspectives on aging and neurodegeneration from psychosocial and neuroscience points of view. How factors including TBI, stroke and alcohol make the brain vulnerable and contribute to neurodegeneration. Clinical overview of Alzheimer's, Parkinson's, Huntington's and ALS and the underlying pathology that differentiates these diseases.

Prerequisite(s): NEUR 2200 or both NEUR 2201 and NEUR 2202.

Lectures three hours a week.

NEUR 3502 [0.5 credit]**Neurodevelopmental Determinants of Mental Health**

Exploration of human brain development, from cell differentiation to complex neural networks. This course considers how variations in neurodevelopment may be associated with conditions such as dyslexia, ADHD, schizophrenia and autism.

Prerequisite(s): NEUR 2200, or both NEUR 2201 and NEUR 2202.

Lectures three hours a week.

NEUR 3999 [0.0 credit]**Co-operative Work Term**

Includes: Experiential Learning Activity

NEUR 4001 [0.5 credit]**Special Topics in Neuroscience**

Each section of NEUR 4001 deals with a different topic. Topics change yearly. Students may register in more than one section of NEUR 4001 but can register in each section only once.

Prerequisite(s): NEUR 3200, or NEUR 3204 and NEUR 3206 and NEUR 3207, or permission of the Department.

Lectures three hours a week.

NEUR 4002 [0.5 credit]**Systematic Reviews and Meta-Analyses**

Introduction to the methods used in conducting systematic reviews and meta-analyses. Topics include: conducting literature searches, extracting relevant literature, assessing quality of studies, synthesizing findings across studies, and the statistical methods used to carry out a meta-analysis.

Includes: Experiential Learning Activity

Precludes additional credit for NEUR 4904.

Prerequisite(s): NEUR 3003 or both NEUR 3001 and NEUR 3002.

Also offered at the graduate level, with different requirements, as NEUR 5203, for which additional credit is precluded.

Lecture three hours a week.

NEUR 4003 [0.5 credit]**Knowledge Mobilization**

Knowledge mobilization concepts, tools, and frameworks, the challenges and value of translational research, and processes involved in integrated knowledge mobilization. Skills to maximize research impacts will be developed. Includes: Experiential Learning Activity
Prerequisite(s): fourth year standing in a Neuroscience program OR permission of the department.
Also offered at the graduate level, with different requirements, as NEUR 5801, for which additional credit is precluded.
Includes: Experiential Learning Activity

NEUR 4200 [0.5 credit]**Seminar on Current Advances in Neuroscience**

Headline research in neuroscience. Topics may include technical and conceptual advances, ethical issues, medical improvement, and social impacts of neuroscience research.
Precludes additional credit for PSYC 4200 (no longer offered).
Prerequisite(s): fourth year standing and one of NEUR 3200, NEUR 3206 or NEUR 3207.
Seminar three hours a week.

NEUR 4202 [0.5 credit]**Seminar on Current Research in Neuroscience and Psychiatric Conditions**

Recent research in clinical neuroscience including biological, developmental, experiential and environmental factors that contribute to mental health conditions. Topics may include depression, schizophrenia, autism, ADHD, eating disorders, sleep-related conditions, substance use disorders, and personality-related conditions.
Prerequisite(s): fourth year standing and one of NEUR 3200, NEUR 3206 or NEUR 3207.
Seminar three hours a week.

NEUR 4203 [0.5 credit]**Seminar on Current Research in Neuroscience and Clinical Neurology**

Recent research in neurological disease, including biological, developmental, experiential and environmental factors that contribute to disease. Topics may include stroke, multiple sclerosis, migraine, seizure disorder, Parkinson's disease, ALS, chronic pain, Alzheimer's disease and concussion.
Prerequisite(s): fourth year standing and one of NEUR 3200, NEUR 3206 or NEUR 3207.
Seminars three hours a week.

NEUR 4301 [0.5 credit]**Neurobiology of Energy Homeostasis**

Focus on neuroanatomical and molecular mechanisms underlying how mammals adapt to changes and challenges in the environment. Topics include regulation of feeding, energy expenditure, water balance, and temperature regulation.
Prerequisite(s): NEUR 3304.
Lectures three hours a week.

NEUR 4302 [0.5 credit]**Sex and the Brain**

Neurobiological processes behind reproductive behaviours in various animal species including humans. Evaluation of data concerning neurobiological differences between sexes, biological determinants of sexual orientation, and relating to neurobiology of sex disorders. Precludes additional credit for NEUR 3302 (no longer offered).
Prerequisite(s): NEUR 3304.
Lectures three hours a week

NEUR 4303 [0.5 credit]**Indigenous Health & Mental Health**

Explores physical and mental health issues in Indigenous communities through cultural, environmental, developmental and biological perspectives, with attention to the impacts of colonialism, intergenerational trauma, and community resilience.
Prerequisite(s): Third year standing or above.
Lectures three hours a week.

NEUR 4305 [0.5 credit]**Immune-Brain Interactions**

Communication between the brain and the immune system; messengers mediating the interaction. How disturbances of immune-brain signaling can lead to disease (multiple sclerosis, Parkinson's) and to changes in mood and cognition.
Precludes additional credit for NEUR 3305 (no longer offered).
Prerequisite(s): NEUR 3200 or NEUR 3207.
Lectures three hours a week.

NEUR 4306 [0.5 credit]**The Neural Basis of Addiction**

Explores how substance use and behavioural addictions impact neural function, contributing to changes in the brain system involved in addiction. Contemporary neurobiological theories of addiction will also be addressed.
Prerequisite(s): NEUR 3204.
Lecture three hours a week.

NEUR 4600 [0.5 credit]**Advanced Lab in Neuroanatomy**

Advanced experiential learning in neuroanatomy, histology and microscopy.

Includes: Experiential Learning Activity

Prerequisite(s): NEUR 3200 or both NEUR 3206 and NEUR 3207, fourth-year standing in a Neuroscience program, a minimum Major CGPA of 9.0 and permission of the Department.

Laboratory/tutorials four hours per week.

NEUR 4801 [0.5 credit]**Neuroethics**

Ethical issues of key importance to current neurobiological research. Topics may include the use of animals in research, stem cell research, genetic diagnosis and gene therapy, neuroimaging, and the effect on identity and autonomy of manipulations such as psychopharmaceuticals and psychosurgery.

Prerequisite(s): NEUR 3200 or both NEUR 3206 and NEUR 3207.

Lectures and seminars three hours a week.

NEUR 4900 [0.5 credit]**Independent Study**

A reading or research course for selected students who wish to investigate a particular topic of interest. Normally students may not offer more than one credit of independent study in their total program.

Includes: Experiential Learning Activity

Prerequisite(s): third- or fourth- year standing and permission of the Department.

NEUR 4904 [1.0 credit]**Honours Research Thesis in Systematic Reviews or Meta-Analyses**

An independent systematic review or meta-analyses undertaken under the direct supervision of a faculty advisor typically from the Department of Neuroscience.

Includes: Experiential Learning Activity

Precludes additional credit for NEUR 4002, NEUR 4905, NEUR 4906, NEUR 4907, NEUR 4908, NEUR 5203.

Prerequisite(s): NEUR 3003 or both NEUR 3001 and NEUR 3002 and fourth-year standing in an Honours Neuroscience program, a minimum Major CGPA of 9.0 and permission of the Department.

Colloquia three hours a week.

NEUR 4905 [1.0 credit]**Honours Workshop**

The course will focus on active learning in areas that include written and oral communication, evaluation and interpretation of results, statistics and data management, emphasizing transferable skills that will be most appropriate for non-research career paths.

Includes: Experiential Learning Activity

Precludes additional credit for NEUR 4906, NEUR 4907 and NEUR 4908.

Prerequisite(s): fourth-year standing in an Honours Neuroscience program and permission of the Department.

Lectures and seminars three hours a week, and colloquia three hours a week.

NEUR 4906 [1.0 credit]**Translational Approach to Indigenous Community Wellness**

This course involves co-developing an Indigenous community-led process or product that addresses a current and specific mental health issue. Involves working in interdisciplinary groups with a community partner.

Includes: Experiential Learning Activity

Also listed as ENSC 4909, ISAP 4909, MPAD 4906.

Precludes additional credit for ENSC 4906, ISAP 4906, ISAP 4907, ISAP 4908, NEUR 4905, NEUR 4907, NEUR 4908.

Prerequisite(s): Fourth-year standing with a minimum Major CGPA of 10.0 and a grade of A- or higher in one of NEUR 3401, NEUR 3402 or NEUR 3403 and permission of instructor. Prior completion of NEUR 4303 recommended.

Seminars or workshops three hours a week. A field trip to the partner community is typically required.

NEUR 4907 [1.0 credit]**Honours Essay**

An independent essay based critical review on a topic in neuroscience, using library resources, under the direct supervision of a Faculty advisor, typically from the Department of Neuroscience. Evaluation is based on a written report.

Includes: Experiential Learning Activity

Precludes additional credit for NEUR 4905, NEUR 4906 and NEUR 4908.

Prerequisite(s): NEUR 3200, or both NEUR 3206 and NEUR 3207, and fourth-year standing in an Honours Neuroscience program, a minimum Major CGPA of 9.0 and permission of the Department.

Colloquia three hours a week.

NEUR 4908 [1.0 credit]
Honours Research Thesis

An independent research project undertaken under the direct supervision of a faculty advisor typically from the Department of Neuroscience. Evaluation is based on a written report and poster.

Includes: Experiential Learning Activity

Precludes additional credit for NEUR 4905, NEUR 4906 and NEUR 4907.

Prerequisite(s): NEUR 3200, or both NEUR 3206 and NEUR 3207, and fourth-year standing in an Honours Neuroscience program, a minimum Major CGPA of 10.0 and permission of the Department.

Colloquia three hours a week.